

The algebra of Furstenberg families

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March 13, 2026

Theorem 1. *If $\mathcal{F}$ is a family, then $\mathcal{F}^*$ is a family. That is, $\cdot^* : \text{Fam}(S) \rightarrow \text{Fam}(S)$.*

Proof. By the definition of family, we need only check that $\mathcal{F}^*$ is an upward closed collection of subsets of S . It is, by definition, a collection of subsets of S . If $A \in \mathcal{F}^*$ and $A \subseteq A' \subseteq S$, to show that $A' \in \mathcal{F}^*$, we need only to show that for all $B \in \mathcal{F}$, $A' \cap B \neq \emptyset$. Since $A \in \mathcal{F}^*$, we have that $\emptyset \neq A \cap B \subseteq A' \cap B$, as desired. $\square$

Theorem 2. *For all families $\mathcal{F}$, $\mathcal{F}^* = \{A \subseteq S \mid S \setminus A \notin \mathcal{F}\}$*

Proof. In the following list, each statement is equivalent to the one that follows it.

1. $A \in \mathcal{F}^*$
 - definition
2. for all $B \in \mathcal{F}$, $A \cap B \neq \emptyset$
 - set theory: for $X, Y \subseteq Z$, $X \cap Y = \emptyset$ iff $Y \subseteq Z \setminus X$
3. for all $B \in \mathcal{F}$, $B \not\subseteq S \setminus A$
 - logic: negation of a universal quantifier
4. not (there exists $B \in \mathcal{F}$, $B \subseteq S \setminus A$)
 - $\mathcal{F}$ is upward closed
5. not ($S \setminus A \in \mathcal{F}$)

$\square$

Theorem 3. *$\cdot^*$ is an involution on the set families: $(\mathcal{F}^*)^* = \mathcal{F}$*

Proof. In the following list, each statement is equivalent to the one that follows it.

1. $A \in (\mathcal{F}^*)^*$
 - theorem 2
2. $S \setminus A \notin \mathcal{F}^*$
 - theorem 2
3. $S \setminus (S \setminus A) \in \mathcal{F}$
 - set theory: for $X \subseteq Z$, $Z \setminus (Z \setminus X) = X$
4. $A \in \mathcal{F}$

$\square$

Theorem 4. *$\mathcal{F}$ is a partition regular family if and only if $\mathcal{F}^*$ is a filter*

Proof. By theorem 1 and theorem 3, $\mathcal{F}$ is family if and only if $\mathcal{F}^*$ is a family. Then, in the following list, each statement is equivalent to the one that follows it.

1. $\mathcal{F}$ is a partition regular family
 - definition
2. for all $A, B \subseteq S$, if $A \cup B \in \mathcal{F}$, then $A \in \mathcal{F}$ or $B \in \mathcal{F}$
 - theorem 3
3. for all $A, B \subseteq S$, if $A \cup B \in (\mathcal{F}^*)^*$, then $A \in (\mathcal{F}^*)^*$ or $B \in (\mathcal{F}^*)^*$
 - theorem 2
4. for all $A, B \subseteq S$, if $S \setminus (A \cup B) \notin \mathcal{F}^*$, then $S \setminus A \notin \mathcal{F}^*$ or $S \setminus B \notin \mathcal{F}^*$
 - set theory: for $X, Y \subseteq Z$, $Z \setminus (X \cup Y) = (Z \setminus X) \cap (Z \setminus Y)$
5. for all $A, B \subseteq S$, if $(S \setminus A) \cap (S \setminus B) \notin \mathcal{F}^*$, then $S \setminus A \notin \mathcal{F}^*$ or $S \setminus B \notin \mathcal{F}^*$
 - logic: “if not p , then not q OR not r ” is equivalent to “if q and r , then p ”
6. for all $A, B \subseteq S$, if $S \setminus A \in \mathcal{F}^*$ and $S \setminus B \in \mathcal{F}^*$, then $(S \setminus A) \cap (S \setminus B) \in \mathcal{F}^*$
 - write $C = S \setminus A$ and $D = S \setminus B$ and use set theory: for $X \subseteq Z$, $Z \setminus (Z \setminus X) = X$
7. for all $C, D \subseteq S$, if $C \in \mathcal{F}^*$ and $D \in \mathcal{F}^*$, then $C \cap D \in \mathcal{F}^*$
 - definition
8. $\mathcal{F}^*$ is a filter

□